Moiré fabrics have characteristic water marked look that is produced during the calendaring process. The effect is developed using either a moiré embossing roller or by a high compression calendaring of two layers of ribbed-base fabric in a single pass. One popular method of preparing moiré fabric features using rollers that have been engraved with a design. The fabric is run between the engraved rollers with some sections of the fabric squeezed to reveal the finished design that has a watery look. This type of application is often used to create material styles that are ideal for evening gowns, clutch handbags, and other types of formal apparel and accessories for women.

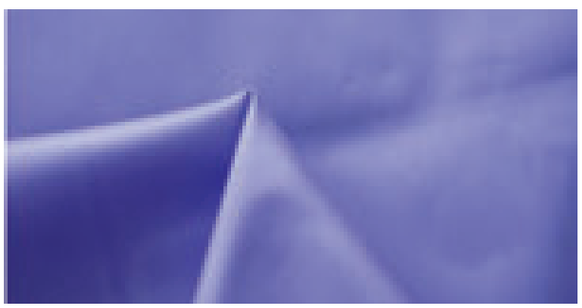


Figure 1.3 Calendared Cotton Fabric

Schreiner finishes on fabrics produce soft lustre and hand by flattening the yarns and surface of a fabric through calendaring. The Schreiner calendar has a metal roller engraved with 200-300 fine diagonal lines per inch that are visible only under a magnifying glass. A Schreiner finish is used on cotton sateen and table damask to make them more lustrous and on nylon tricot to increase its cover.

Fabric embossing can also be carried out in the calendaring machine in which the fabrics can have a pattern imprinted or embossed into it. The embossed pattern is created by passing the fabric between a heated embossing roller and a shaped paper roll. The degree of lustre given to the fabric can be modified by heating the chilled iron roll, by changing the pressure at the nip, by changing the speed of the machine, by carrying the moisture percent present in fabric. Various designs can be created by using a 2-bowl embossed roller or 3-bowl embossed roller arrangement.

1.3.4 Embossing

Embossing is a process that alters the surface of the fabric by providing a three dimensional or raised effect on selected areas. The embossing procedure requires the use of two dyes: one that is raised and one that is recessed. When the dyes are produced, a dyes maker engraves the desired design into several metal plates, which are the embossing dyes for use on an embossing machine. The engraved and the recessed dyes fit into each other so that when the fabric is pressed between them, the raised dye forces the fabric into the recessed dye and creates the embossed impression. A specific level of pressure is applied to the dye in order to squeeze the fibres, which results in a raised area. The embossed design is permanent if the fabric has a thermoplastic fibre content or if a resin is used or the fabric is heat set. Embossing is often used in combination with foil stamping.



Figure 1.4 Fabric Embossing Machine

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